

0 Linear Algebra

0.1 Vectors

A vector is a list of numbers obeying some rules for how to do algebra with them. In math notation a vector is often denoted with either an arrow or boldface

$$\vec{v}$$

$$\mathbf{v}$$

or just v if it is obvious from context that we are talking about a vector.

The **elements** of a vector, $v_1, v_2, \dots, v_n$, are the numbers in the list. The **dimension** of a vector is the number of components. If we write all the elements of the vector on top of each other we have a **column vector**

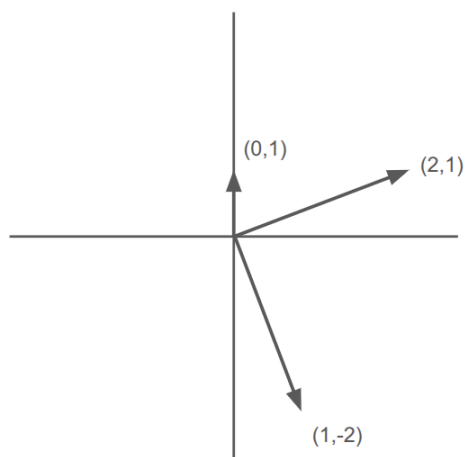
$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

A **row vector** is written by placing all the elements in a row

$$\vec{v}^T = (v_1, v_2, \dots, v_n)$$

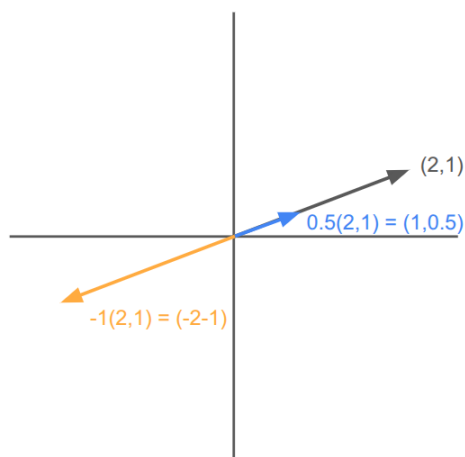
The notation T is an operation called the **transpose**. It means to swap rows with columns. The distinction between rows and columns is important because when we introduce transforms the formulas are different for rows versus columns. The standard convention, which we will follow, is to use columns for vectors.

It is very helpful to have a geometric picture of a vector. We can think of the elements of a vector as co-ordinates of a point and the vector as an arrow based at the origin stretching up to that point. Some examples are shown below



One of the simplest vector operations is scalar multiplication. A **scalar** is just any ordinary number like 2 or -1 or π . Scalar multiplication of a vector is achieved by multiplying every element by the scalar

$$a\vec{v} = \begin{pmatrix} av_1 \\ av_2 \\ \vdots \\ av_n \end{pmatrix}$$



Notice that multiplying by a negative reverses the direction of the arrow.

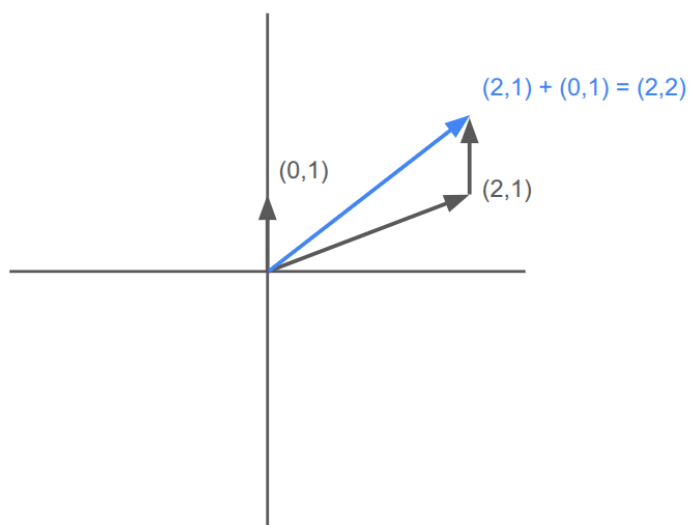
We can add or subtract vectors by adding or subtracting the elements

$$\begin{aligned}\vec{w} &= \vec{u} + \vec{v} \\ \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} &= \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \\ &= \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}\end{aligned}$$

Or we can write the formula for elements of $\vec{w}$

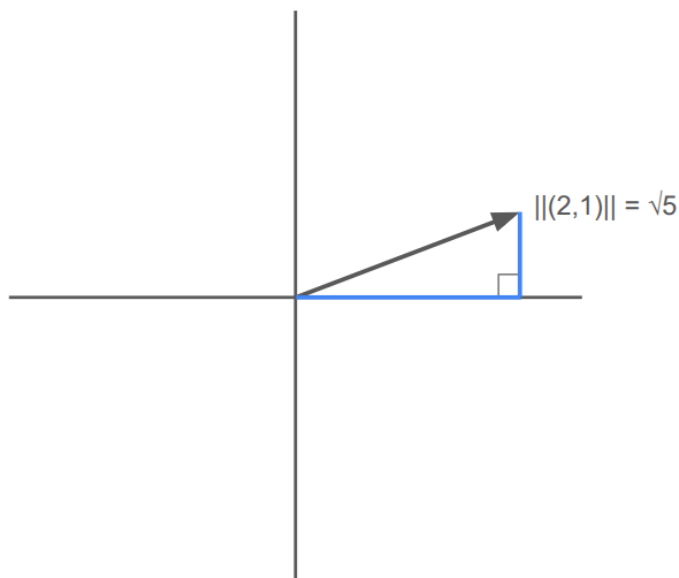
$$w_i = u_i + v_i$$

Geometrically we can think of putting the vectors arrow to tail



Any vector has a norm or **length** given by Pythagoras' theorem

$$||\vec{v}|| = \sqrt{\sum_{i=1}^n v_i^2}$$

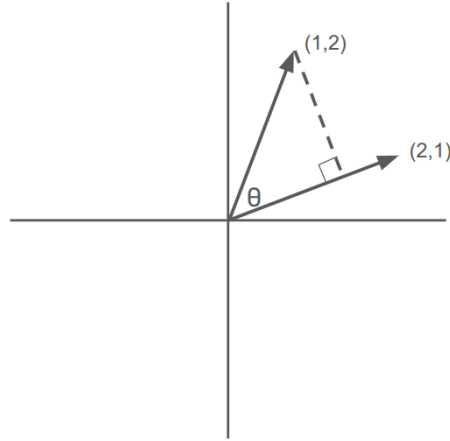


The **dot product** combines two vectors into a scalar by multiplying the elements and adding them.

$$\vec{v} \cdot \vec{w} = \vec{v}^T \vec{w} = \sum_{i=1}^n v_i w_i$$

The terms $\vec{v} \cdot \vec{w}$ and $\vec{v}^T \vec{w}$ are equivalent ways to denote the dot product. The second form will make more sense when we discuss matrix multiplication but shows you can think of the dot product as a way to multiply row vectors by column vectors. The norm can be written as a dot product of a vector with itself

$$\|\vec{v}\|^2 = \vec{v} \cdot \vec{v}$$



The geometric interpretation requires you to do some trigonometry, but shows that the dot product tells us about the **angle** between vectors

$$\vec{x} \cdot \vec{y} = \|\vec{x}\| \|\vec{y}\| \cos \theta$$

0.2 Matrices

Matrices are simply rectangular arrays of numbers

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

We will denote a matrix with a capital letter: $A, B, C, \dots$ and

$$A_{ij}$$

means the i^{th} row and the j^{th} column. It is valid, and often helpful, to think of a matrix as a ‘row of column vectors’

$$A = (\vec{c}_1 \quad \vec{c}_2 \quad \vec{c}_3 \quad \dots)$$

or a ‘column of row vectors’

$$A = \begin{pmatrix} \vec{r}_1 \\ \vec{r}_2 \\ \vec{r}_3 \\ \vdots \end{pmatrix}$$

A matrix with N rows and M columns is called an $M \times N$ matrix. If the number of rows and columns are equal, $N = M$, then the matrix is said to

be **square**. If $N = 1$ we have a column vector, and if $M = 1$ we have a row vector.

Matrices can be added and subtracted element-wise e.g.

$$A_{ij} + B_{ij} = C_{ij}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 6 & 9 \end{pmatrix}$$

To add and subtract matrices they have to have the same shape! We can also multiply a matrix by a number by multiplying all its elements

$$B_{ij} = cA_{ij}$$

We can multiply a (column) vector by a matrix using the dot product. The vector $\vec{w}$ that we get by multiplying v by A has components

$$w_i = \sum_j^M A_{ij}v_j$$

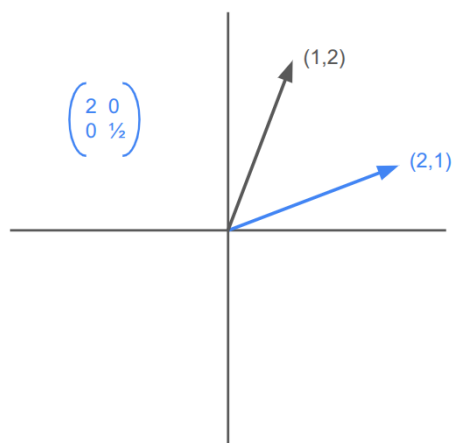
Which is equivalent to taking the dot product of each row with the vector

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1a + 2b + 3c \\ 4a + 5b + 6c \end{pmatrix}$$

We can think of a matrix acting on a vector as rotating and scaling the vector. For example the matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

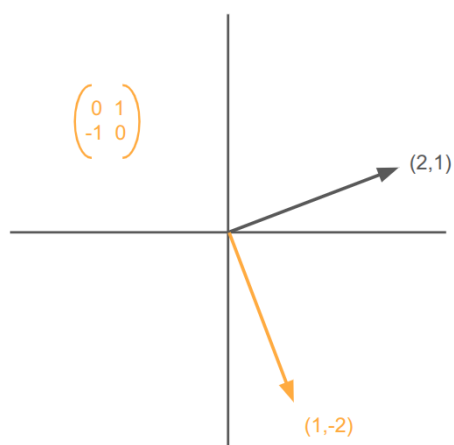
stretches the x-component by a factor 2 and shrinks the y-component by a factor $\frac{1}{2}$.



While the matrix

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Rotates a vector by 90 degrees clockwise.



And in general a matrix multiplication does some mixture of both rotation and stretching.

The special matrix with all ones on the diagonal

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

returns the original vector unchanged and is called the **identity matrix**.

$$I\vec{x} = \vec{x}$$

Any matrix with only non-zero values on the main diagonal like this is called a **diagonal matrix**

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

To multiply two matrices together we repeat the matrix vector multiplication algorithm on each column of the right hand matrix. Each component of the product matrix is given by

$$C_{ij} = \sum_k^K A_{ik} B_{kj}$$

i.e. the dot product of the i^{th} row with the j^{th} column. When multiplying matrices their shapes have to match. We can only multiply a $N \times M$ matrix with a $M \times Q$ matrix, the inner dimensions must match. The result will be an $N \times Q$ matrix.

With matrix multiplication the order matters! Multiplying A by B is not the same as multiplying B by A

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 7 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} -2 & -2 \\ 4 & 6 \end{pmatrix}$$

This property is called **non-commutativity**. An exception is diagonal matrices, which commute with any other matrix. On the other hand matrix multiplication is always **associative**

$$A(BC) = (AB)C$$

The transpose of a matrix means swapping the rows and columns

$$(A^T)_{ij} = A_{ji}$$

i.e. the i^{th} row and j^{th} column of the matrix A^T is equal to the j^{th} row and

i^{th} column of A .

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

The transpose of an $N \times M$ matrix is an $M \times N$ matrix. A matrix which is equal to its own transpose is called **symmetric**. Symmetric matrices are common in applications, for example, the adjacency matrix of a network is symmetric. Transposing swaps the order of a product

$$\begin{aligned} (AB)_{ij}^T &= AB_{ji} = \sum_k A_{jk} B_{ki} \\ &= \sum_k B_{ki} A_{jk} \\ &= \sum_k (B^T)_{ik} (A^T)_{kj} \\ &= (B^T A^T)_{ij} \end{aligned}$$

i.e.

$$(AB)^T = B^T A^T$$

A **linear equation** is an equation like

$$A\vec{x} = \vec{b}$$

Where A and $\vec{b}$ are known and we want to find the vector $\vec{x}$ which is transformed into $\vec{b}$ by A . This kind of equation arises frequently in applications, for example, solving this equation is how a straight line is fit to a scatter plot. If we knew a matrix A^{-1} with the property that

$$A^{-1}A = I$$

then we could multiply both sides on the left by A^{-1}

$$\begin{aligned} A^{-1}A\vec{x} &= A^{-1}\vec{b} \\ I\vec{x} &= A^{-1}\vec{b} \\ \vec{x} &= A^{-1}\vec{b} \end{aligned}$$

giving the unknown vector $\vec{x}$. The matrix A^{-1} is called the **inverse** of A . In practical problems we don't directly find the inverse, but it is useful to know that one exists.

0.3 Vector Spaces

Vectors live in a **vector space** which is often some very high dimensional space, though we will usually do the drawings in 2d. For vectors representing some data, a very important idea is to break down the vector into sums of other vectors.

A set of vectors is said to be **linearly dependent** if at least one of the vectors can be written as a linear combination of the others. A linear combination means adding, subtracting and multiplying by scalars. For example in the set of three vectors

$$\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$$

the $\vec{v}_3 = 0.5\vec{v}_1 + 0.5\vec{v}_2$, therefore this set is linearly dependent. A set is **linearly independent** if it isn't linearly dependent.

The vectors $\vec{v}$ and $\vec{w}$ are **orthogonal** if

$$\vec{v}^T \vec{w} = 0$$

Using the geometric interpretation of the dot product, you can show this means the angle between the two vectors is 90° . If every vector in a set is orthogonal to every other this is an **orthogonal set**. If the vectors also have unit length i.e.

$$\|\vec{v}\| = 1$$

then the set is called **orthonormal**. We can **normalize** a vector by dividing it by its length

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$

You can easily show that $\|\hat{v}\| = 1$

A **basis** for an N dimensional vector space is a linearly independent set of N vectors. Given these vectors, we can build any other vector in the space from linear combinations of the basis vectors. If the basis is orthonormal it is particularly easy to work with. The **standard basis** is

$$\vec{e}_{j_i} = \delta_{ij}$$

in two dimensions

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

This is not the only basis though. For example,

$$\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

is also an orthonormal basis. One of the most common things we want to do is find a basis that suits our data.

When we have vectors $\vec{x}$ in the basis $\vec{e}_i$ this means we know the coefficients c_i in

$$\vec{x} = \sum_i c_i \vec{e}_i$$

If we want to transform to a new basis $\vec{v}_i$ that means we want to find the coefficients d_j in

$$\vec{x} = \sum_j d_j \vec{v}_j$$

To do this we express the new basis vectors in terms of the old ones

$$\vec{v}_j = \sum_i A_{ij} \vec{e}_i$$

where A_{ij} is the i^{th} component of the j^{th} new basis vector in the old basis. The matrix with these numbers as elements is exactly what we need to change the basis. To see this

$$\begin{aligned} \vec{x} &= \sum_j d_j \vec{v}_j \\ &= \sum_j d_j \sum_i A_{ij} \vec{e}_i \\ &= \sum_i \left(\sum_j A_{ij} d_j \right) \vec{e}_i \end{aligned}$$

This means

$$c_i = \sum_j A_{ij} d_j$$

or

$$\vec{x}_{old basis} = A \vec{x}_{new basis}$$

where $\vec{x}_{old basis}$ is the list of the old components c_i and $\vec{x}_{new basis}$ is the list of the new components d_j .

If the bases are orthonormal then if $j \neq k$

$$\vec{e}_k^T \vec{v}_j = \sum_i A_{ij} \vec{e}_k^T \vec{e}_i = 0$$

$$\vec{e}_j^T \vec{v}_j = \sum_i A_{ij} \vec{e}_i^T \vec{e}_i = A_{jj}$$

since $\vec{v}_j$ is normalised we must have $A_{jj} = 1$, with a bit of thought this implies $A^{-1} = A^T$ so

$$A^T \vec{x}_{old\ basis} = \vec{x}_{new\ basis}$$

0.4 Eigenvectors

If our data is encoded in a matrix then there is a basis associated with every matrix that is often very helpful in solving problems related to that matrix. This is called the **eigenbasis** which consists of the matrix's **eigenvectors** which are associated with particular **eigenvalues**.

As we discussed, matrices can be thought of as transforming a vector in some way, rotating it as well as stretching and squashing it along each dimension. However there are some special vectors which are not changed much by the matrix. For these vectors all the matrix does is make them longer or shorter, it doesn't rotate or distort them. These vectors satisfy the **eigenvalue equation**

$$A\vec{v}_i = \lambda_i \vec{v}_i$$

λ_i is the eigenvalue and $\vec{v}_i$ the eigenvector. Sometimes the same eigenvalue is associated with two different eigenvectors. We will ignore this possibility, see any book on linear algebra for more details.

Eigenvectors are quite hard to find for large matrices, we discuss one method in Chapter 5. But it is often useful to simply know they exist for theoretical work. If we found all the eigenvectors we could put them in a matrix as columns

$$V = (\vec{v}_1 \quad \vec{v}_2 \quad \dots \quad \vec{v}_N)$$

Then

$$AV = \Lambda V$$

Where

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$$

is a diagonal matrix with the eigenvalues along the diagonal. We can manipulate this equation by multiplying *on the left* by the inverse of V

$$\begin{aligned} AV &= \Lambda V \\ V^{-1}AV &= V^{-1}\Lambda V \end{aligned}$$

As diagonal matrices commute with any other matrix,

$$\begin{aligned} V^{-1}AV &= \Lambda V^{-1}V \\ V^{-1}AV &= \Lambda I \\ V^{-1}AV &= \Lambda \end{aligned}$$

This transformation of the matrix A is said to **diagonalize** it. What it basically means is that if we write all vectors in terms of the eigenvectors of A then the action of A is very simple, it just scales the coordinates by an amount proportional to the eigenvalues. In particular, it does not *mix up* different components, so instead of solving one huge complicated interdependent problem, we can solve N independent ones.

If A is symmetric (as it will always be in this course) the eigenvectors and eigenvalues are especially nice. In general eigenvalues can be complex (involving factors of $\sqrt{-1}$), but for symmetric matrices they are always real. The eigenvectors are also orthogonal. To see this take two different eigenvectors of A , $\vec{v}_1$ and $\vec{v}_2$, with different eigenvalues. We have

$$\begin{aligned} A\vec{v}_1 &= \lambda_1\vec{v}_1 \\ A\vec{v}_2 &= \lambda_2\vec{v}_2 \end{aligned}$$

Taking dot products with the other vector gives

$$\vec{v}_1^T A\vec{v}_2 = \lambda_2 \vec{v}_1^T \vec{v}_2$$

we also have

$$\begin{aligned} \vec{v}_2^T A\vec{v}_1 &= \lambda_1 \vec{v}_2^T \vec{v}_1 \\ (\vec{v}_2^T A\vec{v}_1)^T &= \lambda_1 (\vec{v}_2^T \vec{v}_1)^T \\ \vec{v}_1^T A^T \vec{v}_2 &= \lambda_1 \vec{v}_1^T \vec{v}_2 \\ \vec{v}_1^T A\vec{v}_2 &= \lambda_1 \vec{v}_1^T \vec{v}_2 \end{aligned}$$

Subtracting these equations gives

$$(\lambda_2 - \lambda_1)\vec{v}_1^T \vec{v}_2 = 0$$

We chose different eigenvectors, so $\lambda_2 \neq \lambda_1$ and therefore $\vec{v}_1^T \vec{v}_2 = 0$. This means **eigenvectors of a symmetric matrix are orthogonal**. Also if $\vec{v}$ is an eigenvector, so is $\alpha\vec{v}$ for some number α , we can always choose eigenvectors to have norm 1, therefore the basis is orthonormal. You can now show that $V^{-1} = V^T$ and

$$V^T A V = \Lambda$$

It is *much* easier to transpose a matrix than take its inverse, so this is a tremendous simplification for symmetric matrices, which is why we like them so much!

For one example of how useful this is, we will sometimes have vectors which vary over time

$$\vec{x}(t)$$

and equations like

$$\frac{d\vec{x}}{dt} = A\vec{x}$$

This means as $\vec{v}$ changes over time its components get mixed together by the matrix A . However, if we change to the eigenbasis then

$$\begin{aligned}\frac{d\vec{x}}{dt} &= V \Lambda V^T \vec{x} \\ V^T \frac{d\vec{x}}{dt} &= \Lambda (V^T \vec{x}) \\ \frac{d\vec{x}'}{dt} &= \Lambda \vec{x}'\end{aligned}$$

This equation for $V^T \vec{x} = \vec{x}'$ is much simpler, the components don't get mixed up at all and we have N *independent* equations.